



www.efy.in

EFYGROUP
YOURS SINCE 1969

Branding is
what people
say about
you when
you are not
in the room.



-Jeff Bezos



Go to System Options
Select Wireless LAN
Configure your Wi-Fi connection
Install the remaining softwares.
First go to a virtual environment.

```
$> python -m venv /path/to/myenv
$> source myenv/bin/activate
(myenv) bera@raspberrypi:~$ pip install
-r requirements.txt
```

Coding

Configure the paths for image storage, the output number-plate directory, and the machine-learning model files for both vehicle detection



Fig. 4: Recognised cars and their numberplates



Fig. 5: Author's prototype used for testing

and number-plate recognition. Keep the model files in the same folder as the code or specify the correct paths within the script (refer to Fig. 1, code snippet lines 25 and 26).

Save the file as lpr10_pi5.py or another preferred name. Fig. 2 shows the code path configuration.

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to
<https://sourcecode.electronicshobby.com>
to access and download source code
and other resources for this project

Testing

Now open the terminal and run Python followed by the code file, as mentioned earlier:

```
(myenv) bera@raspberrypi:~$ python
lpr10_pi5.py
(myenv) bera@raspberrypi:~$ python lpr3_
pi5.py sample.png
```

The first command runs using the live camera feed. The second command analyses an image file and prints the detected licence plate number.

The live script also creates a log file named detection_log.txt, which

records recognised number plates along with timestamps. Fig. 4 shows the terminal output showing successful license plate detection from a sample image using the Python LPR script on Raspberry Pi.

The lpr10_pi5.py script uses a two-step verification process (yolov8n.pt and kbest.pt). A vehicle must first be detected before the number plate is processed.

The lpr3_pi5.py script uses only the kbest.pt model and

assumes that a vehicle is already present in the supplied image.

The TFT touchscreen, keyboard, and mouse are optional. Once deployed at the operational level, these peripherals can be removed. The code is not suitable for use with the Hailo-8L AI HAT.

Caution. Lighting conditions affect recognition accuracy. Images that appear clear during the day may become unclear at night. Adequate spot lighting significantly improves reliability. **EFY**



Somnath Bera is an electronics enthusiast. He is a freelancer and has presented several articles across the globe.